



ELS10: Python Dictionaries - Study Notes

Definition • Role • Real-life Examples



A. Python Dictionaries

Definition: A **dictionary** in Python is a collection of **key-value pairs**. Each key is unique and used to access its corresponding value.

```
student = {"name": "Alice", "age": 18, "course": "ICT"}
```

Role: Used to store related data in key-value form, making it organized and accessible.

Real-life Example: Like a real dictionary — you look up a **word (key)** to find its **meaning (value)**.



B. Access Items

Definition: Accessing items means getting the value of a specific key.

```
student = {"name": "Alice", "age": 18}
print(student["name"])    # Output: Alice
```

Role: Retrieve stored information quickly by key name.

Real-life Example: Looking up someone's **phone number** in your **contacts** using their name.



C. Change Items

Definition: Change the value of an existing key.

```
student["age"] = 20
```

Role: Used when data updates over time, such as new age or score.

Real-life Example: Updating a student's age or email address in a school database.

D. Add Items

Definition: Add a new key-value pair to an existing dictionary.

```
student["grade"] = "A"
```

Role: Expands data when new information becomes available.

Real-life Example: Adding a new contact or phone number to your contact list.

E. Remove Items

Definition: Delete one or more key-value pairs.

```
student.pop("age")    # Removes the item with key "age"
```

Role: Cleans up or manages memory by removing unnecessary data.

Real-life Example: Removing a discontinued product from a shop's stock list.

F. Loop Items

Definition: Use a loop to go through all keys, values, or both.

```
for key, value in student.items():  
    print(key, ":", value)
```

Role: Processes all data in a dictionary, like displaying or checking them.

Real-life Example: Displaying all students and their marks in a report.



G. Copy Items

Definition: Make a duplicate of a dictionary without affecting the original.

```
copy_student = student.copy()
```

Role: Prevents accidental changes to the original data when you need a backup.

Real-life Example: Making a photocopy of a record before editing the original.



H. Nested Dictionaries

Definition: A dictionary inside another dictionary.

```
school = {  
    "student1": {"name": "Alice", "age": 18},  
    "student2": {"name": "Bob", "age": 19}  
}
```

Role: Organizes complex data structures like databases.

Real-life Example: A school database with multiple students and their details.



I. Dictionary Methods

Definition: Built-in functions that help manage dictionary operations.

Method	Description
<code>dict.keys()</code>	Returns all keys
<code>dict.values()</code>	Returns all values
<code>dict.items()</code>	Returns key-value pairs
<code>dict.get(key)</code>	Returns value safely
<code>dict.pop(key)</code>	Removes item by key
<code>dict.clear()</code>	Removes all items

Role: Make dictionary handling faster, simpler, and safer.

Real-life Example: Like using shortcuts or tools in Microsoft Word to speed up editing.